

optimization could reflect a reluctance to take risk on the part of beneficiaries, the firm, or both.⁴²

This is easier said than done. Several liability measures are computed by actuaries; the two most common are the Accumulated Benefit Obligation (ABO) and the Projected Benefit Obligation (PBO). The ABO is the value of the benefits currently earned by employees and retirees. This is the contractual liability if the firm shut down immediately. Healthy firms, however, are going concerns, and so the PBO also counts the future expected salaries of current employees. (Thus the PBO is larger than the ABO.) But even the PBO is incomplete: firms generally grow over time, and the PBO does not count the future benefits of new hires. ERISA's measure of liabilities is the ABO. The true economic liability valuation probably exceeds even the PBO.

Public pension plan liabilities are hugely underestimated because the valuation methods assume that the pensions are very risky, while in reality state pension benefits are close to risk free.⁴³ The pension benefits should therefore be valued, like safe government bonds, using low discount rates, rather than with the high discount rates implied by risky equities. To see the problem with using high discount rates, consider a couple with a mortgage.⁴⁴ This is a liability, and its value does not change whether they hold equities or bonds in their retirement accounts. Yet in public pension plans, their actuarial liabilities magically become smaller if they hold more equities in their 401(k). This is pure fiction in economic terms, of course, because the mortgage liability is not affected by the rate the couple earns in their 401(k) plan.

It also matters whether the ABO or the PBO is used.⁴⁵ If the fund's liabilities are measured by the ABO, then the best *liability-hedging* portfolio is a portfolio of bonds with the same cash flows or *duration* (the average time the liability outflows come due). This implies the optimal asset allocation for a pension fund

⁴² ERISA Section 404(a)(1) specifies that investment fiduciaries must perform their duties solely in the interest of participants and for the exclusive purpose of providing participants with retirement benefits. This is part of the *prudent man* investment standards set by ERISA. Thus, under ERISA, the surplus utility function cannot represent the utility function of the fund manager.

⁴³ In many states, pension liabilities are guaranteed under state constitutions. New York State, for example, gives state pension liabilities the same seniority as general obligation debt. (See "Who Watches the Watchman? New York State Common Retirement Fund," Columbia CaseWorks ID#110307.) Thus, pension liabilities are close to risk free and should be discounted using discount rates resembling risk-free (Treasury or municipal) bond yields. Under accounting standards (GASB 25) and actuarial standards (ASOP 27), the discount rate is much higher, and is the expected long-term return on assets. This effectively considers public pension liabilities to be extremely risky. It also violates a fundamental principle in economics that the value of any stream of payments should be independent of the way that it is financed, which is the Miller and Modigliani (1958) principle. (Merton Miller and Franco Modigliani were awarded Nobel Prizes in 1990 and 1985, respectively.) See also Novy-Marx and Rauh (2009, 2011).

⁴⁴ This argument is in Ang and Green (2011).

⁴⁵ See Black (1989) for more details on these arguments.